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Perspectives (Systematic Literature Review)

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The effectiveness of digital sexual health interventions for young adults: a systematic literature review (2010–2020)

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Summary

There has been a proliferation of digital sexual health interventions targeting adolescents; however, limited evaluative reviews have compared the effectiveness of multiple digital tools for sexual health literacy and behaviour change. This study conducted a systematic literature review, screened 9881 records and analysed 61 studies. Findings suggest that websites and mobile phones dominate digital sexual health interventions, with a majority effectively delivering cognitive (e.g. awareness and attitudes about sexual and reproductive health) and behavioural outcomes (e.g. abstinence and use of contraception). The most popular sexual health promotion mechanisms were interactive websites, text messaging and phone calls, and online education programmes, followed by mobile applications—fewer studies in this review utilized social media, games and multimedia. Previous reviews focused on single outcome measures (e.g. sexually transmitted infection testing) to assess interventions' effectiveness. The current review moves beyond single outcome measures to cover a wider range of behavioural and non-behavioural sexual health issues and contexts covered in the literature. Four main categories were analysed as outcomes: cognitive perceptions, promoting sexual health-related behaviours, promoting sexual health-related products and services, and impact (viral load). Seventy-nine per cent of interventions focused on preventive sexual health behaviours and products (e.g. condoms) and services (e.g. HIV testing). Overall, 75% of studies effectively changed sexual health behaviour and cognitive perceptions. However, the digital-only tools did not vary from the blended formats, in influence outcomes, even after categorizing them into behavioural or non-behavioural outcomes. Compared to previous systematic reviews, more studies from the last decade used rigorous research design in the form of randomized controlled trials, non-randomized control trials, and quasi-experiments and lasted longer.

Lay summary

Sexual health literacy among adolescents is essential to avert unwanted pregnancies or abortions and sexually transmitted diseases. Culturally sensitive and age-appropriate interventions are required to educate youth about safe-sex practices and increase their self-efficacy and accessibility to health products and services. Meta-analytical and systematic literature reviews have identified limitations in traditional interventions that rely heavily on classroom-based activities and lectures to educate adolescents about safe sex. In recent years, digital tools have proved productive; however, limited evaluative reviews have compared the effectiveness of multiple digital technologies (such as smartphones, web-based programs, social media, games and multimedia) used within sexual health interventions. This systematic review assessed 9881 articles and synthesized 61 experimental studies on adolescent sexual behaviour conducted in the last decade to identify tools that positively influence cognitive and safe-sex behaviour among youth. Interactive websites, mobile technology and online education programmes were popular promotional tools, but very few studies utilized multimedia alongside games and social media. Findings from this review could facilitate future research and practice in adolescent sexual health.

Keywords: systematic literature review, sexual health interventions, young adults, digital technology, behaviour change

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INTRODUCTION

Adolescents represent one-sixth of the global population (World Health Organization [WHO], 2020a, b). This group undergoes a transition from childhood to adulthood, facing unique health challenges. Most adolescent mortality and morbidity are caused by risk-related behaviour, such as sexual intercourse without contraceptives or unprotected intercourse or activities with multiple partners which can result in the contraction of sexually transmitted infections (STIs) and unwanted pregnancy (WHO, 2020a, b). Each year about 333 million young adults are diagnosed with an STI worldwide (Dehne and Riedner, 2005). Furthermore, the WHO reported that 1 million girls under the age of 15 years and 12 million under 19 years give birth, signalling a severe risk of unintended pregnancies (WHO, 2020a, b). To overcome such challenges and improve health and well-being, adolescents require protection from harm and support to make independent and reliable decisions. WHO and the United Nations International Children's Emergency Fund (UNICEF) encourage evidence-based sexual health interventions to increase knowledge around sexual health and promote safe-sex practices.

One of the main barriers young adults face in seeking sexual health is their inability to gain accurate and trustworthy sexual health information and services. This is due to restrictive policies, teachers', and parents' reluctance to openly discuss sexual health issues, parental control, limited income, and lack of confidentiality (WHO, 2020a, b). Some of these barriers have been overcome recently by the advent of digital media. Program designers in the past decade (Salam *et al.*, 2016) have turned their attention to digital and computer-based sexual health interventions where access is unrestricted, the cost to the user is low and confidentiality is ensured. To deliver sexual health-related information to young adults, such interventions usually employ websites (e.g. Brady *et al.*, 2015), mobile applications (e.g. Jeong *et al.*, 2017), social media platforms (e.g. Fisser, 2013), text messaging (e.g. Rana *et al.*, 2015) and video games (e.g., Christensen *et al.*, 2013).

Literature indicates that digital sexual health interventions deliver several advantages. First, it can reach at-risk or marginalized populations without compromising their privacy and safety. It also increases cost-effectiveness and efficiency for service providers with limited budgets for facilitator training and materials (DeSmet *et al.*, 2015; Brayboy *et al.*, 2018). Second, interactive computer-based interventions increase user engagement and deliver prompt feedback (Wayal *et al.*, 2014; Shafii *et al.*, 2019), are less threatening to patients, and provide greater anonymity to users than face-to-face modes (Marsch *et al.*, 2015). Third, in countries where sexual issues are taboo (Marsch *et al.*, 2015; Dehghani *et al.*, 2019), web-based strategies are highly relevant in providing user anonymity and widespread dissemination of sexual health information to youth at low costs. Not surprisingly, digital interventions deliver information (for instance, about contraceptives and STIs more effectively than face-to-face interventions (DeSmet *et al.*, 2015).

The aims of this systematic review study are threefold. First, to build on previous reviews (Salam *et al.*, 2016) and reflect the current trends and practices of the safe-sexual health promotion sector when digitizing interventions. As the past few years experienced an exponential increase in adolescents' access and use of digital and social media and the expansion of platforms where interactions occur between young adults, peers and health services, an update of previous reviews is paramount. Second, this review analyses current digital sexual health interventions' effectiveness and presents insights into effective digital interventions in the sexual health field. There is no recent systematic review that looks beyond specific digital interventions (e.g. serious games: DeSmet *et al.*, 2015), scope (Salam *et al.*, 2016), age (e.g. 13–24: Guse *et al.*, 2012) and sexual health outcomes (e.g. STI screening: Daher *et al.*, 2017) to understand the effectiveness of digital sexual health interventions. Third, this review compares the digital-only interventions with those that combine digital and non-digital tools. An integrated model of digital and classroom-based or face-to-face interventions (also known as blended learning) is gaining popularity (Coyle *et al.*, 2019). However, limited rigorous evaluative reviews have assessed such sexual health interventions (Decker *et al.*, 2020). These insights will highlight the strengths and weaknesses of contemporary digital interventions and provide opportunities for future interventions.

METHOD

The systematic literature review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Moher *et al.*, 2009). Following PRISMA protocols, this review sought to identify studies that evaluated technology-based sexual health programmes published in peer-reviewed journal articles to answer three research questions:

RQ1: What insights can we draw from the past 10 years of digital sexual health interventions targeting young adults?
RQ2: What contributes to effective digital sexual health interventions for young adults?
RQ3: Do non-digital features enhance programme delivery?

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Literature and database search

A two-step search process was used for this review. In step one, index terms (i.e. thesaurus and subject headings) were adapted to suit different databases. A search of Web of Science, Pubmed and Scopus was undertaken to analyse text words in the title, abstract and keywords. Following this analysis, the final keywords related to technology-based sexual health were selected, and the search strategy was developed. In the second step, a comprehensive electronic search was performed in eight electronic bibliographic databases, including Web of Science, Scopus, CINAHL Complete, Cochrane, Embase, ProQuest Central, Medline and PsycINFO. Each database was searched by combining terms from each of four conceptual categories: digital (computer*, digital media, social media, Internet, technology, video gam*, gaming, games, cell* phone*, mobile, smartphone, electronic* mail, e-mail, e-mail, hypermedia, web, website*, blog*, weblog*, software*, text messag*, SMS, short messag* service, MMS); sexuality education or sexual health (sexual behavio*, sex behavio*, sex* education, sexual knowledge, sexually transmitted disease*, sexually transmitted infection*, HIV, condom*, contraception, safe sex, unsafe sex, unprotected sex, protected sex, sexual abstinence, unplanned pregnancy, unwanted pregnancy, sexual activit*, sexual health, family planning); adolescent (adolescen*, teen*, young adult*, youth) and intervention (intervention*, campaign*, program*, trial*, experiment*, school-based, community health*, initiative*, education*, promot*, prevent*). The keywords were customized to each database-specific indexing term (e.g. MeSH in PubMed). The search was conducted during the second 2 weeks of August 2020.

Selection criteria and study screening

Articles retrieved from the database search were stored in EndNote v.X9.2. Duplicate studies, pre-2010 articles and publications in the non-English language were excluded from the Endnote library by the first author. The remaining articles were then randomly assigned to four independent reviewers. Each reviewer screened the title and abstracts of studies to exclude irrelevant articles. When it was difficult to determine an article's relevance based on its abstract, the full paper was retrieved and examined by two reviewers independently to decide its eligibility.

The following inclusion criteria were applied during full-text screening:

1. A specific time parameter (2010–20) was selected for this systematic review to study trends in digital sexual health education interventions' adoption and efficacy. Articles that were dated outside this period were excluded from the study.
2. Empirical studies that focused on sexual health promotion/education. Articles were excluded if they presented only qualitative findings. Clinical studies that discussed clinical procedures were excluded from this study
3. Included a digital intervention. Digital interventions, sometimes referred to as eHealth interventions, are programmes that utilize information technology to promote, prevent or maintain healthy behaviour (Montanaro *et al.*, 2015). Articles were excluded if they did not integrate digital technology such as smartphones, computers, tablets, multimedia and social media.
4. Targeted young people. WHO (2020b) describes *young people* as aged 10–24. *Adolescents* refer to individuals aged 10–19, while *youth are* represented by the 15–24 age group (WHO, 2020b). Studies were excluded when it became apparent that none of the participants fit this criterion.

Intercoder reliability was calculated between two reviewers independently assessing a specific quota of full texts. For each group comprising of two reviewers, the Cohen's Kappa score revealed a high agreement level: Group 1 (author 1 and 5; n = 23) = 0.86; Group 2 (author 1 and 2; n = 30) = 0.88; Group 3 (author 2 and 3; n = 19) = 0.75 and Group 4 (author 3 and 5; n = 8) = 0.7). Disagreements between investigators were resolved through rigorous discussions.

Search outcomes

The details of the procedures involved in the article selection process are outlined in Figure 1.

The authors identified 16 826 records through the database search. After removing articles pre-dated 2010 (n = 3486) and duplicate records (n = 3459), a total of 9881 unique records were identified. We applied the inclusion criteria (which is mentioned in the previous section) to screen the title and abstracts of 9881 articles. At this stage, the authors identified additional duplicate records (n = 79), non-English articles (n = 21), non-journal articles (n = 460), non–sexual health articles (n = 490) and irrelevant articles (n = 8611) that were outside the inclusion criteria. This resulted in 220 eligible records. We screened the full text of 220 articles to determine further eligibility. One hundred and fifty-nine (n = 159) records were removed because they were non-behaviour focused, non-empirical, qualitative, non-intervention or non-adolescent focused. No further articles were identified through backward and forward searching—the final set comprised 61 eligible studies with an experimental design.

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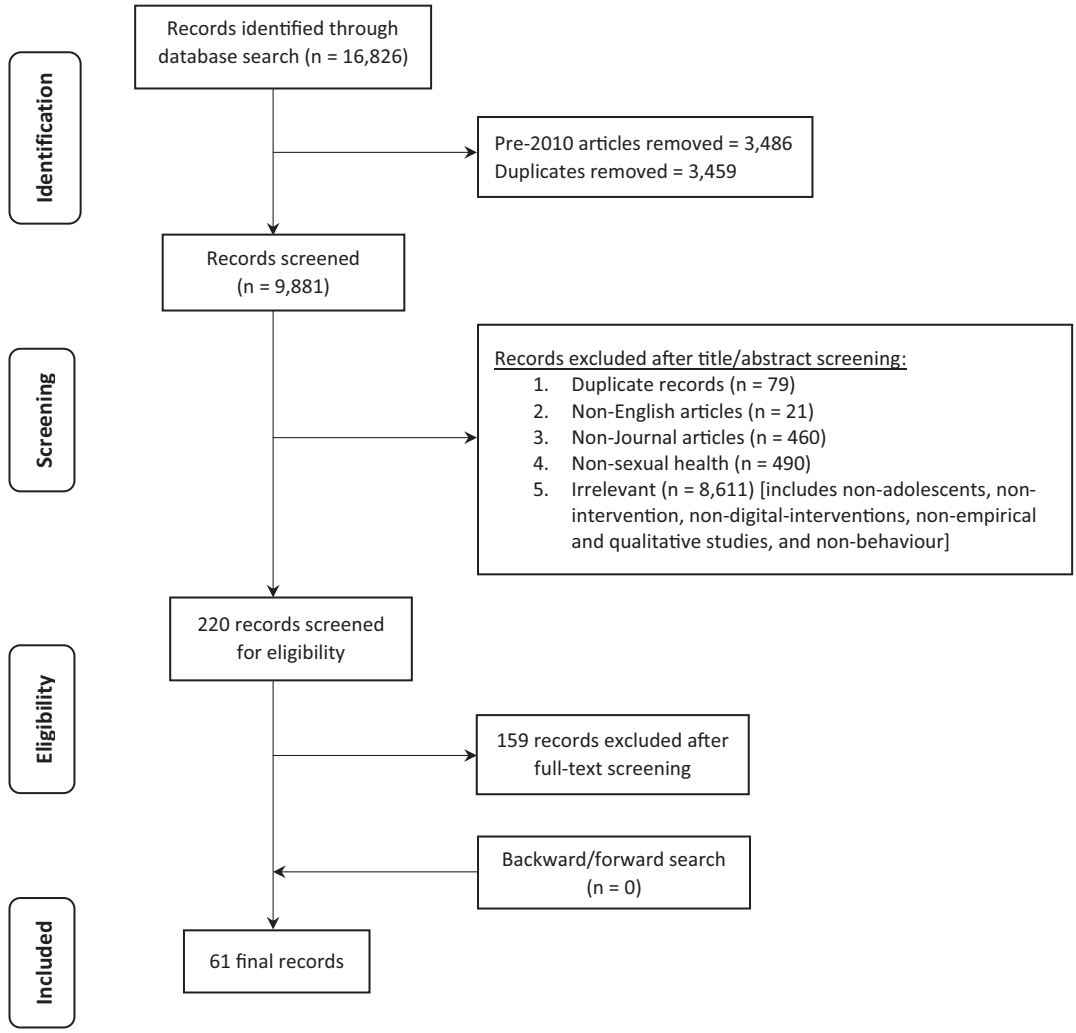


Fig. 1. PRISMA flowchart.

Data extraction and analysis

A meta-analysis was not possible in this study due to the heterogeneity of the tested interventions and study populations and the diversity in outcomes metrics. Not all studies reported precise effect sizes and odds ratios, thus, limiting our analysis to a qualitative description of outcomes reported by 61 studies. The authors used a piloted matrix to extract data from the included articles to assess study characteristics and evidence synthesis. Extracted data consisted of the first author's surname, year of publication, stated purpose of the study, theories informing intervention design, study setting, study design, participant characteristics, sample size, data collection methods, primary variable(s) of interest, descriptions of the interventions, features of effective interventions, description of outcomes and study limitations. Authors analysed extracted data based on location, sample size, age, target audience, intervention details, duration, digital and non-digital components, behavioural focus, behavioural outcomes and effectiveness.

RESULTS

In total, 61 studies representing 59 unique interventions qualified for inclusion. Supplementary Tables 1 and 2 provide information about the interventions, duration, target audience, sample size, outcomes and intervention effectiveness. Table 1 provides a summary of the outcomes.

Description of included studies

More than half of the studies were from the USA (*n* = 32), followed by the UK, Australia and Nigeria,

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Table 1. Summary of outcomes

		Cognitive outcomes	Behavioural outcomes	Products/service outcomes	Impact
		Not effective studies, Effective studies[1]			
Digital tools	Mobile text and phone calls	1,7[2] 0,0[3]	2,4 5,11	7,7	0,2
	Mobile apps	1,4 1,3	1,2 1,2	10	0,0
	Web- and online-based	0,13 0,4	1,4 2,10	1,8	0,1
	Online education programmes	2,8 0,2	4,5 3,7	2,4	0,0
	Social media	0,1 0,0	0,1 1,1	1,1	0,0
	Games	0,5 0,2	2,0 2,1	0,1	0,1
	Multimedia	0,3 0,1	1,1 0,2	0,1	0,0
Digital tools versus blended	Digital only	1,27 1,8	6,12 8,23	8,14	0,2
	Combination of digital and non-digital	3,14 0,4	5,5 6,11	4,8	0,2
Duration of studies	Short (13 and fewer weeks)	2,24 1,10	5,10 6,15	3,10	0,2
	Long (14 and longer weeks)	1,14 0,1	5,6 7,17	9,11	0,2

[1]Effective at $p < 0.05$.
[2]All studies that measured this particular outcome and reported a level of effectiveness. The total exceeds 61, as most studies measured more than one outcome.
[3]Studies where this outcome was the most advanced measure (according to the persuasion process, behavioural outcomes are more advanced than cognitive outcomes). Only one outcome (the most advanced one) was assigned to each study. Behaviour, its tangible form (product/service), and impact were considered equall y advanced. Thus, the total is equal to 61.

with three studies from each country (e.g. Gold *et al.*, 2011; Bailey *et al.*, 2013; John *et al.*, 2016). Mexico, Sweden and South Africa contributed two studies each (e.g. Gottvall *et al.*, 2010; Doubova *et al.*, 2017; Hacking *et al.*, 2019). Only one study was found for Argentina (Stankievich *et al.*, 2018), Chile (Villegas *et al.*, 2015), Colombia (Chong *et al.*, 2020), Ghana, Hong Kong (Chu *et al.*, 2015), Iran (Dehghani *et al.*, 2019), the Netherlands (Mevissen *et al.*, 2011), Portugal (Carvalho *et al.*, 2016), Senegal (Massey *et al.*, 2009), South Korea (Jeong *et al.*, 2017), Spain (Ballester-Arnal *et al.*, 2015), Tajikistan (McCarthy *et al.*, 2020), Tanzania (Haruna *et al.*, 2018) and Uganda (Linnemayr *et al.*, 2017). Most interventions (n = 38) targeted young adults aged 15–24 years and adolescents aged 10–19 years (n = 22), while only one intervention, though targeted at youth, had unspecified age. Few studies were gender specific. For instance, 12 studies targeted young female adults; 7 studies were designed for males, while 42 studies were generic.

Digital tools used

Our review suggests that websites and mobile phones dominate digital sexual health interventions, with a majority effectively delivering cognitive and behavioural outcomes. The behaviours were either promoted directly (e.g. abstinence) or in a tangible form (e.g. condoms). The most popular mechanisms for sexual health promotion were interactive websites (n = 16), text messaging and phone calls (n = 16), and online education programmes (n = 12), followed by mobile applications (apps) (n = 7). Fewer studies in our review

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utilized social media (n = 2), games (n = 5) and multimedia (n = 3). Social media and multimedia-based interventions used platforms such as Facebook, TV, radio, videos and digital storytelling to promote sexual health among adolescents.

Websites and online education programmes

Sixteen studies in this review employed websites. A higher proportion of website-based interventions were found to improve sexual health outcomes in adolescents, evidenced by an increase in knowledge (Danielson *et al.*, 2016; Doubova *et al.*, 2017), access to sexual health services (Mortimer *et al.*, 2015; Brown *et al.*, 2016), increase in STI prevention practices (Massey *et al.* 2009; Mevissen *et al.* 2011; Spielberg *et al.* 2014; Ballester-Arnal *et al.* 2015; Mortimer *et al.* 2015; Lustria *et al.* 2016; Doubova *et al.* 2017; Horvath *et al.* 2017), reduction in risky sexual behaviours (Villegas *et al.*, 2015), increase in condom use efficacy (Starling *et al.*, 2014) and increase in ART adherence (Naar-King *et al.*, 2013).

Twelve studies in this review used online education programmes to disseminate information about safe sex and STI risk reduction (e.g. Marsch *et al.*, 2015; Klein *et al.*, 2017; Kaufman *et al.*, 2018; Scull *et al.*, 2018; Scull *et al.*, 2019; Chong *et al.*, 2020). The findings from this category produced mixed results. While some studies (n = 8) found online programmes to be effective in increasing knowledge, four studies (n = 4) found contradicting results (e.g. Castillo-Arcos *et al.*, 2016; Klein *et al.*, 2017; Peskin *et al.*, 2019; Shafii *et al.*, 2019). For example, two computer-delivered interventions by Carvalho *et al.* (2016) positively influenced condom use amongst adolescents; however, internet-based sessions cited by Castillo-Arcos *et al.* (2016), Chong *et al.* (2020) and Shafii *et al.* (2019) were unsuccessful in delivering behavioural outcomes. On the other hand, online programmes that targeted STI prevention, such as Marsch *et al.* (2015) and Kaufman *et al.* (2018), recorded positive results.

Mobile applications

Seven studies employed mobile applications, and the majority were found to be effective in increasing sexual health knowledge (Brayboy *et al.*, 2017; Mesheriakova *et al.*, 2017; Dehghani *et al.*, 2019), STI prevention (Jeong *et al.*, 2017), pregnancy prevention (Manlove *et al.*, 2020) and condom use (Jones *et al.*, 2013). Two studies reported insignificant results (Brayboy *et al.*, 2017; Nielsen *et al.*, 2019). Given the sensitivity surrounding open discussions on sexual health, it is not surprising that users value privacy, security and credibility. Online statistics show that in 2020, almost 50% of people worldwide reported opening an app more than 10 times a day and millennial over 50 times a day (Iqbal, 2020). This indicates the popularity and favourability of mobile applications among adolescents.

Text messaging and phone calls

A high proportion of studies (n = 16) in this review engaged with adolescents via text messages and phone calls. Eleven studies delivered positive outcomes in terms of influencing HIV knowledge and perceptions (Gold *et al.*, 2011; Merrill *et al.*, 2018), increasing condom use and HIV testing (Yao *et al.*, 2018), increasing ART and PrEP adherence (Belzer *et al.*, 2014; Stankievich *et al.*, 2018; Hacking *et al.*, 2019; Liu *et al.*, 2019), increasing Chlyamadia testing (McCarthy *et al.*, 2016), and enhancing self-care among HIV+ adolescents (John *et al.*, 2016). Five studies found text messaging to be ineffective in increasing the use of contraception (Gold *et al.*, 2011; Juzang *et al.*, 2011; Lim *et al.*, 2012; McCarthy *et al.*, 2016; Ybarra *et al.*, 2018), increasing ART adherence among HIV-positive adolescents (Linnemayr *et al.*, 2017), and rising sexual and reproductive health knowledge (Rokicki *et al.*, 2017; Ybarra *et al.*, 2018). Conflicting results are evident in the literature regarding SMS effectiveness. For example, while Belzer *et al.* (2014) found text messaging to increase ART adherence among HIV-positive adolescents, Linnemayr *et al.* (2017) found the tool ineffective. Similarly, when targeting HIV knowledge and prevention, Merrill *et al.* (2018) found text messaging useful, while Juzang *et al.* (2011) did not.

The popularity of text messaging as a clinic-based tool to communicate with HIV+ youth and young adults is evident in this review. In general, text messaging dominates interventions in sexual health education of adolescents and youth, with one in every four interventions using text messaging and phone calls. This can be attributed to the effective results such tools yield and the favourability and usability among adolescents. Text messaging is also preferred for its ease of implementation for large groups and different contexts, making it a highly usable and attractive intervention platform. Over 35% of 13–17-year olds rated texting as their number one communication method (Marketing Charts, 2018). Text messages were also found to serve multiple sexual health issues. For example, Rokicki *et al.* (2017) found messages to reduce unwanted pregnancies effectively. In a different context, Stankievich *et al.* (2018) found text messaging to increase adherence to ART medications and decrease HIV Viral Load (VL). Hence, text messaging is an attractive mechanism for health practitioners for its applicability across diverse sexual health issues.

Social media

Fewer studies (n = 2) in this review utilized social media channels for sexual health promotion. Interventions

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that disseminated STI prevention messages via Facebook page and encouraged ART adherence through Facebook-based support groups produced mixed outcomes, confirming previous review findings by Guse *et al.* (2012). In our review, Dulli *et al.*'s (2020) study found that social media interventions only affect HIV knowledge and do not influence ART adherence behaviour. On the other hand, Bull *et al.* (2012) found the Facebook page to influence condom perceptions and use effectively, but the intervention was conducted for a short term. Bull *et al.* (2012) argue that social media could be a useful tool to reach a large number of youth audiences. However, the impact of that reach is limited as this age group is not known to engage with health organizations on Facebook.

Games

Interactive games, on the other hand, were employed scarcely in the identified studies (n = 5). Three game-based interventions produced positive results; two did not (e.g. Christensen *et al.*, 2013; Fiellin *et al.*, 2017). Game-based learning proved efficacious in influencing knowledge and attitudes towards sex (Chu *et al.*, 2015; Haruna *et al.*, 2018), reducing shame and decreasing risky sexual behaviours (Christensen *et al.*, 2013), and increasing ART knowledge and adherence among HIV+ youth (Whitely *et al.*, 2018). Games were ineffective in delaying sexual initiation (Fiellin *et al.*, 2017). In 2019, 34.5 billion games were accessed, downloaded and played by online users (Iqbal, 2020). In 2020, the most common web-based applications were gaming applications, highlighting its popularity amongst millennial (Deshdeep, 2020). Sexual health educators and health practitioners are slowly catching on to the gaming world; however, applications remain limited in numbers.

Multimedia

Limited studies in our review adopted multimedia tools. All three studies that employed multimedia tools generated positive results. Different types of mass media were used, including radio and television (Sznitman *et al.*, 2011; Solorio *et al.*, 2016). The limited number of studies in this category can be attributed to the availability of more interactive tools that offer a two-way communication approach (e.g. text messaging and applications). Although the number of people who have access to TV and radio worldwide has increased, the daily usage rates have declined, indicating that such platforms have limited reach (Richter, 2020). Adolescents are still watching TV but in a different way. Statistics show that young audiences are likelier to use streaming services such as Netflix and Amazon Prime Video than traditional TV (Richter, 2020). This presents an opportunity for sexual health educators to utilize new platforms targeting young adults (e.g. TV shows on Netflix that discuss sexual health problems).

Combination of digital and non-digital tools

Twenty-one out of 61 studies used blended mode (digital and non-digital tools) to deliver sexual health interventions to adolescents. Fifteen of these studies reported effectiveness in achieving their outcomes. Studies adopting digital-only tools reflected a higher effectiveness ratio than the blended tool interventions, both cognitive and behavioural outcomes, but the statistical difference was not significant. Mobile-based interventions incorporated traditional features such as clinical visits (Linnemayr *et al.*, 2017), face-to-face interactions with health professionals (Belzer *et al.*, 2014), PrEP and sexual reproductive health education by a health educator or a nurse (Jeong *et al.*, 2017; Rokicki *et al.*, 2017; Liu *et al.*, 2019), and a free hotline number to gain additional information about reproductive health from health professionals (Rokicki *et al.*, 2017). Four out of six studies were effective.

Web-based interventions (e.g. websites and online education programmes) were supplemented by classroom-based discussions and role-plays (Gottvall *et al.*, 2010; Ballester-Arnal *et al.*, 2015), print-based information materials and contraceptives (Gottvall *et al.*, 2010), at-home STD testing by collecting specimen and mailing to the lab for analysis (Lustria *et al.*, 2016), and STI treatment at a pharmacy or at a clinic (Spielberg *et al.*, 2014). Online education programmes were also supplemented by face-to-face classroom-based information sessions, presentations and role-plays (Castillo-Arcos *et al.*, 2016; Klein *et al.*, 2017; Kaufman *et al.*, 2018; Markham *et al.*, 2020), and parent–child homework activities (Peskin *et al.*, 2019). Seven out of 11 studies were effective.

Interventions that used social media (such as Facebook-based support groups) also provided in-person support (Dulli *et al.*, 2020). Only one intervention combined both mobile and electronic technology with traditional communication tools. Solorio *et al.* (2016) used social media outreach, a website, a reminder system via text messaging, radio advertisements, print materials and a toll-free hotline number to encourage HIV testing among the target audience.

Intervention outcomes

From the 61 studies representing 59 digital sexual health interventions, 61 outcomes were assessed. We categorized the outcomes into four groups: (i) cognitive perceptions assessing likability of stimuli, creating awareness and influencing attitudes; (ii) promotion of preventative sexual health behaviours; (iii) promotion of products and services and (iv) impact. Forty-five studies evaluated cognitive perceptions,

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28 interventions evaluated preventive sexual health behaviours, 34 studies evaluated the uptake of sexual health products (e.g. condoms) and services (e.g., HIV testing), and 4 studies reported an impact measure (VL). Forty-five studies attempted to enhance cognitive perception; 91% of the results were effective, while among 28 studies promoting sexual health behaviours, 60.7% were effective. Promoting products and services was effective in 64.7% of the 34 studies. All four studies reported achieving impact successfully.

Twenty-one interventions targeted STI/HIV prevention, including screening/testing and increasing knowledge, attitudes and intentions towards preventative behaviour. Furthermore, condom use was a primary outcome of eight studies and is often considered STI/HIV preventive behaviour. This predominant focus on STIs and HIV is unsurprising as they are major global health issues, with approximately 1 million people acquiring STIs daily (WHO, 2020a, b). We have observed that digital sexual health prevention interventions focus heavily on STI and HIV prevention efforts and ignore other aspects of sexual health-related issues such as sexuality, consensual sex, intimate partner violence and substance abuse, implying a gap exists in terms of comprehensive sexual health interventions for adolescents and young adults.

Duration

The mean duration of 61 studies was 25 weeks (SD = 40.11), with a median of 13 weeks, ranging from 1 day to 5 years. One could interpret 25 weeks as a sufficiently long duration of an intervention to achieve intermediate outcomes such as knowledge and attitudes. Duration length was not related to intervention effectiveness (χ^2 (9, N = 57) = 16.76, p = 0.05). In terms of digital tools, shorter and longer duration interventions utilize a higher proportion of web and online tools and mobile tools, respectively.

DISCUSSION

RQ1: What insights can be drawn from the last 10 years of digital interventions?

To the authors' knowledge, the last comprehensive systematic review of adolescent-based sexual and reproductive health interventions was conducted by Salam *et al.* (2016), identified gaps in the comparability of results and suggested using more rigorous study designs and longer term follow-ups, and the need for equitable outcomes. This current evaluative review synthesized empirical studies published between 2010 and 2020, out of which half (n = 30) were published in the last 3 years (2017–2020). The majority of the studies that we reviewed used rigorous research design in the form of randomized controlled trials (n = 40), non-randomized control trials (n = 13), and quasi-experiments (n = 6), indicating an increase in the volume of high-standard scientific studies in the area of adolescent sexual health. Another notable characteristic of the pool of studies within our systematic review was intervention length, which averaged 25 weeks. Interventions conducted for a longer period allow practitioners to evaluate sustained behavioural outcomes, as recommended by previous systematic reviews (e.g. Salam *et al.*, 2016).

RQ2: Intervention effectiveness of digital tools

Overall, 75% of studies found digital interventions to yield an effective change in sexual health perceptions and behaviour. Web and online interventions contributed to the highest number of effective interventions (50%), followed by mobile text and phone calls (23.9%), mobile apps (10.9%) and social media, multimedia, and games (15.7%). Twenty-five per cent found digital tools to be ineffective. Mobile interventions contributed to the highest percentage of ineffective interventions (46.67%), followed by web and online (33%) and social media, multimedia, and games (20%). A chi-square test of independence showed an insignificant association between the employed digital tools and the effectiveness of the interventions χ^2 (6, N = 61) = 4.02, p = 0.67. These findings highlight the need to analyse other factors that may significantly affect intervention effectiveness (e.g. study design, segmentation, theory base).

RQ3: Effectiveness of digital-only interventions versus blended mode (a combination of digital and non-digital tools)

Fifteen interventions that used the blended approach produced positive results in influencing knowledge, self-efficacy, perceptions and safe-sex behaviour, while six blended interventions failed to deliver. On the other hand, 31 digital-only interventions were effective, and 9 were not. While a higher proportion of digital-only studies were effective than the blended ones, this difference was not statistically significant (χ^2 [3, N = 61] = 1.04, p = 0.79). This finding is consistent with previous reviews on blended learning for sexual health education, indicating mixed results (Coyle *et al.*, 2019; Decker *et al.*, 2020). More research is required to ascertain whether digital technologies can reinforce traditional sexual health advocacy and behaviour change promotion methods among young adults (Decker *et al.*, 2020).

Limitations and future research

Several important limitations apply to the current study. First, the study is limited by the search

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parameters and search terms. For example, the review only includes studies that empirically test the effectiveness of digital sexual health interventions through quantitative analysis that have been published in peer-reviewed English literature. Hence, studies that evaluate qualitatively (e.g. focus groups) or solely non-digital interventions (e.g. classroom-based) or published in non-English and non–peer reviewed studies were excluded. A specific limitation arises from the parameters of the current review where qualitative studies are excluded. Qualitative studies can advance understanding of effectiveness through in-depth interviews and focus groups with sexual health practitioners and youth groups. Future systematic reviews may include other data collection methods (e.g. interviews) and additional data collection strategies or studies that evaluate interventions for other age groups (e.g. children) to widen the understanding of sexual health interventions. Second, due to the heterogeneity of the tested interventions, study populations and reporting of results, a meta-analysis was not possible; this study could only provide a qualitative description of outcomes. Only a limited number of the included studies provided clear effect sizes and odds ratios, limiting comparing of intervention effectiveness. Third, while the current review provides insights into digital sexual health interventions' effectiveness on a variety of sexual health outcomes, analysis of specific digital tools' effectiveness (e.g. apps) on specific targeted behaviours (e.g. condom use) was not conducted, warranting the need for future research to understand the efficacy of each digital tool.

Fourth, even though our study explored how different media interact with each other to influence audience behaviour, an in-depth understanding is required of the various persuasion techniques and transmedia employed by the interventions. The current study did not highlight these points as they were beyond the study scope. Future research should analyse both these aspects and provide new insights to improve the effectiveness of digital sexual health interventions.

Similarly, analysis of specific target audiences (e.g. Hispanic youth) and their acceptance and receptiveness of specific digital interventions may yield interesting findings for practitioners in specific cultures and communities. Finally, a comparison with other approaches (e.g. face-to-face lectures) was not possible by focusing on digital interventions. Future research can benefit from comparing digital interventions to traditional approaches to recommend the best approach practitioners should utilize.

Future research should examine digital sexual health interventions for adolescents by comparing the behaviour change effectiveness of single outcome interventions (e.g. condom use) with holistic interventions (as recommended by the WHO, 2020a, b) that additionally promote the prevention of violence, enhance understanding of sexuality, increase STI knowledge and promote risk-minimizing behaviours. This review also noted the absence of studies reporting on Return of Investment (ROI). While a study of sexual health intervention effectiveness is critical to understanding the power of digital strategies, so is efficiency (Schäferhoff *et al*., 2020); as funding agencies increasingly require reporting of ROI, future research should measure and report this metric.

As a limited number of digital sexual health interventions employed games, multimedia and social media platforms, we could not conclude their ability to deliver effective behaviour change. Social media is yet fully utilized and implemented as a sexual health education tool, and future research should mainly analyse their efficacy. This confirms previous systematic review findings, where social media interventions were limited in numbers (Guse, 2012). This is surprising, given that over half of the world's population uses social media, especially the youth audience (Chaffey, 2020). In 2018, 70% of youth aged 13–17 reported using social media more than once daily. Furthermore, game-based interventions are gaining steady interest in other public health areas (e.g. alcohol; Rodriguez *et al*., 2014). They are less utilized in interventions targeting sexual health education (DeSmet *et al*., 2015).

Future systematic literature reviews should evaluate interventions based on social marketing benchmarks. Past systematic literature reviews have employed Andreasen's (2002), and National Social Marketing Centre's (2020) benchmarks to assess the extent of the presence of social marketing benchmarks in a variety of behaviour change interventions (e.g. the review of nutrition interventions by Carins *et al*. [2014] and physical activity interventions by Fujihira *et al*. [2015]). However, such a study is lacking in digital sexual education interventions. To highlight a few benchmarks, limited systematic reviews have explored the effects of customizing digital interventions for adolescents and whether programme efficacy increases through a co-design approach. Previous studies, such as Xia *et al*. (2016), have revealed a positive association between the presence of benchmarks and the intervention's success in promoting physical activity. A similar association test will help identify critical benchmarks that influence safe-sex behaviours among adolescents and guide future practices.

CONCLUSION

The current review aimed to extend the understanding of digital sexual health intervention effectiveness when delivered to adolescents. Furthermore, we aimed

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to highlight the trends, opportunities and limitations of the past 10 years of research in the area. Our review supports digital interventions to improve adolescents' sexual health in concordance with existing reviews. There is a need to test digital interventions beyond websites, mobile applications and text messaging to understand the effectiveness of video games, social media and multimedia interventions in achieving positive sexual health outcomes. Our findings indicate that most digital sexual health interventions aimed at adolescents' focus on one or complementary behavioural outcomes (e.g. HIV prevention and contraception use) with limited attention to a holistic sexual health education that includes a broad range of topics.

Supplementary material

Supplementary material is available at *Health Promotion International* online.

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